

ASSESSMENT TOOL-3

Application Based Questions

1. Auto-Scaling and Load Balancing During Seasonal Sales

An e-commerce company may experience a sudden increase in users during events such as **Black Friday, Diwali sales, or Amazon-style flash sales**. If the company uses fixed server capacity, the servers may become overloaded and cause slow response or downtime.

Auto-Scaling

Cloud auto-scaling automatically adjusts the number of servers according to workload.

For example:

- Normal traffic → 5 servers may be sufficient.
- During a major sale → traffic increases sharply, so auto-scaling adds more servers.
- After the sale → extra servers are removed.

This ensures that resources are available when required without continuously paying for maximum capacity.

Load Balancing

A **load balancer** distributes incoming customer requests among available servers.

For example:

Customers → Load Balancer → Server 1 / Server 2 / Server 3 / Server 4

If one server becomes overloaded or unavailable, traffic can be redirected to other healthy servers.

Reducing Downtime and Improving User Experience

The combination of auto-scaling and load balancing:

- Prevents server overload.
- Reduces application response time.
- Improves availability.
- Handles sudden traffic increases.
- Prevents a single server failure from affecting all users.

Customers can therefore browse products and complete payments with fewer interruptions.

Role of Monitoring Tools

Monitoring tools continuously track:

- CPU and memory usage.

- Number of active users.
- Response time.
- Error rates.
- Network traffic.
- Server health.

When resource usage reaches a predefined threshold, monitoring systems can trigger auto-scaling or generate alerts for administrators.

Real-World Example

During a **Diwali online shopping sale**, millions of customers may simultaneously search for products and place orders. The load balancer distributes requests across multiple servers, while auto-scaling adds additional servers when traffic increases. After the sale, the additional servers are automatically removed.

Conclusion:

Auto-scaling and load balancing provide flexible capacity, reduce downtime, control costs, and provide a smoother shopping experience during unpredictable traffic spikes.

2. Secure Cloud Storage for Backup Data

Organizations often need to store large amounts of backup data, including databases, application files, and system backups.

Cloud storage provides durability, redundancy, security, and flexible pricing.

Durability and Data Redundancy

Cloud storage providers store multiple copies of data across different physical systems or locations.

If one storage device fails, another copy can be used.

For example:

Original Data → Copy 1 → Copy 2 → Copy 3

This protects data against hardware failures and other infrastructure problems.

Organizations can also use **cross-region replication** for additional protection against regional disasters.

Encryption

Backup data can contain sensitive information, so encryption should be used.

Encryption at rest:

Protects data while it is stored.

Encryption in transit:

Protects data while it is being transferred between systems.

Encryption keys should be securely managed and access to them should be restricted.

Access Control

Access control ensures that only authorized users can access backup data.

The organization can use:

- Role-Based Access Control (RBAC).
- Multi-factor authentication.
- Least-privilege permissions.
- Strong identity management.
- Audit logs.

For example, a normal employee should not have permission to delete production backups.

Cost Advantages

Compared with traditional storage systems, cloud storage can:

- Avoid large upfront hardware investments.
- Use pay-as-you-go pricing.
- Automatically scale storage capacity.
- Reduce maintenance costs.
- Provide different storage classes for frequently and rarely accessed backups.

Old backups can be moved to lower-cost archival storage.

Disaster Recovery Example

Suppose an organization's primary database is accidentally deleted due to a system error.

The company can:

1. Identify the latest valid backup.
2. Restore the backup to a new environment.
3. Recover the database.
4. Redirect applications to the recovered database.

If the primary datacenter is unavailable, a replicated backup in another region can be used for recovery.

Conclusion:

Cloud storage provides durable and redundant backup infrastructure while encryption and access control protect sensitive information. It also reduces the need for expensive physical storage infrastructure.

3. CDN for Global Mobile Applications

A mobile application with users around the world needs to deliver content quickly. A **Content Delivery Network (CDN)** improves performance by storing frequently requested content on geographically distributed edge servers.

How CDN Improves Performance

Without a CDN:

User → Main Cloud Server

With a CDN:

User → Nearest Edge Server → Content

The edge server may already have a cached copy of the requested content.

This reduces the distance that data needs to travel and therefore reduces latency.

Role of Edge Servers

Edge servers are located in different geographical locations closer to users.

For example:

- User in Chennai → Chennai/nearby edge location.
- User in London → European edge location.
- User in Singapore → Asian edge location.

Instead of every request traveling to a central server, cached content can be delivered from a nearby location.

Benefits

CDNs can:

- Reduce latency.
- Improve page/app response time.
- Reduce load on origin servers.
- Handle large numbers of simultaneous users.
- Improve availability.
- Provide smoother video or game content delivery.

Cloud Global Availability

Cloud providers operate infrastructure across multiple geographic regions and availability zones. Applications can be deployed across multiple regions, while CDNs distribute content closer to users.

Real-Time Streaming Example

Consider a **live cricket streaming application**.

During a major match, millions of viewers may request the same video content. The CDN caches and distributes video segments through edge locations.

Viewer → Nearest Edge Server → Video Segment

This reduces pressure on the central streaming server and helps users receive video with lower latency.

Gaming Example

For an online multiplayer game, CDN edge locations can quickly deliver:

- Game updates.
- Game assets.
- Patches.
- Static content.

This reduces download time and improves the overall user experience.

Conclusion:

CDNs and edge servers bring content closer to users, reducing latency and improving performance. They are especially useful for global applications such as streaming platforms and online games.

4. Cloud Security and Compliance for a Financial Institution

Financial institutions handle highly sensitive information such as customer accounts, payment details, and transaction records. Therefore, cloud adoption must satisfy strict security and regulatory requirements.

Regulatory Compliance

Cloud providers offer security and compliance capabilities that can help organizations meet regulatory requirements.

They may provide:

- Compliance certifications and reports.
- Security controls.
- Audit support.
- Identity management.
- Encryption services.
- Logging and monitoring.

However, using a compliant cloud service does not automatically make the customer's entire application compliant. The financial institution must configure and operate its environment correctly.

Logging and Monitoring

All important activities should be logged, including:

- User login attempts.
- Database access.
- Changes to security settings.
- Financial transactions.
- Administrative actions.

Monitoring systems analyze these logs and can identify suspicious activity.

For example, repeated failed login attempts or unusual transaction activity can generate security alerts.

Encryption

Financial data should be encrypted both:

At rest:

When stored in databases, backups, or storage systems.

In transit:

When moving between applications, users, and cloud services.

Strong key management should be implemented, with restricted access to encryption keys.

Access Control

The institution should use:

- Multi-factor authentication.
- Role-Based Access Control.
- Least-privilege permissions.
- Strong identity verification.

Only authorized employees and applications should access financial information.

Data Sovereignty

Data sovereignty means that data may need to remain within specific geographic or legal boundaries.

The organization can ensure this by:

- Selecting cloud regions that meet legal requirements.
- Restricting where sensitive data is stored.
- Applying geographic policies.
- Monitoring data movement.
- Using encryption and appropriate contractual controls.

Example: Online Banking Transaction

Consider a customer transferring ₹50,000 through online banking.

The transaction should:

1. Be authenticated using strong authentication.
2. Travel through an encrypted connection.
3. Be processed by authorized services.
4. Be stored in an encrypted database.
5. Generate an audit log.
6. Be monitored for suspicious behavior.

If an unauthorized person attempts to modify the transaction, security monitoring can detect abnormal activity.

Conclusion:

Financial institutions can use cloud security controls, encryption, identity management, logging, monitoring, and appropriate geographic controls to protect sensitive financial transactions and support regulatory compliance.

5. Containers, Virtual Machines, and Kubernetes

Containers and virtual machines both provide isolation for applications, but they work differently.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Virtualization	OS-level	Hardware-level
Operating System	Share host kernel	Each VM has its own OS
Startup	Very fast	Comparatively slower
Resource usage	Low	Higher
Size	Usually smaller	Usually larger
Portability	Very high	Also portable, but heavier
Best suited for	Microservices	Complete OS/application environments

A container packages an application along with its dependencies while sharing the host operating system kernel. Because containers do not require a separate full operating system for each application, they generally consume fewer resources.

Kubernetes Orchestration

When an application contains hundreds of containers, manually managing them becomes difficult. **Kubernetes** automates container management.

It can handle:

- Container deployment.
- Scheduling.
- Service discovery.
- Health monitoring.
- Automatic restart.
- Scaling.
- Rolling updates.
- Load distribution.

Scaling Example

Suppose an online shopping application's payment service normally runs with 3 containers.

During a sale:

3 containers → High traffic → Kubernetes adds more containers → 10 containers

After traffic decreases, unnecessary containers can be removed.

Portability Across Clouds

Containers package the application and its dependencies in a consistent format. The same container image can be deployed on different cloud platforms with suitable orchestration and configuration.

For example:

Developer Laptop → Cloud A → Cloud B → On-Premises

This reduces dependency on a single cloud environment and makes application migration easier.

Microservices Example

Consider an **online food delivery application** divided into multiple microservices:

- User Service
- Restaurant Service
- Order Service
- Payment Service
- Delivery Service
- Notification Service

Each service can run in its own container.

If the Order Service experiences heavy traffic, only the Order Service can be scaled without unnecessarily scaling the other services.

Conclusion:

Containers provide lightweight and portable application environments, while Kubernetes simplifies the deployment, management, and scaling of large numbers of containers. This makes containerization particularly suitable for modern **microservices-based cloud applications.**